

A CASE STUDY IN AUTORESEARCH

From Black Box to White Box

Measuring how an instruction file shapes an autonomous research agent's behavior

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Motivation

WHY THIS MATTERS

An autoresearch loop is a language model running its own research cycle: read instructions, propose an experiment, run it, interpret, decide what's next — repeating without a person in the loop.

What we observe

Change the instruction file — `program.md` — and the agent's research *process* changes: how many experiments it runs, how broadly it searches, how it stops.

What we don't know

Which specific sentence caused which specific change, and by how much. The relationship is observed, never explained.

Why that gap is costly

Without it we cannot **predict** behavior under an untried instruction, and we cannot **explain** behavior to someone else in a way they can act on. “It behaved differently because I changed the instructions” is not actionable; “it ran fewer experiments because the stopping rule was stricter, which explains half the variation” is.

The approach: borrow from AutoML

SAME PROBLEM, ALREADY SOLVED ONCE

AutoML already has tools for exactly this kind of question — just applied to hyperparameters instead of instructions. An instruction file is, structurally, no different: a set of discrete, largely independent choices that jointly determine behavior.

1	2	3
Ablation design	Surrogate model	Causal validation
A small, principled subset of configurations to run — not all of them, not random ones.	An interpretable model: which axis explains how much of the behavior.	Predict a held-out configuration's behavior, then check it for real.

Step 2 — five instructional axes

TURNING PROGRAM.MD INTO A CONFIGURATION VECTOR

Every possible `program.md` is generated from a 5-bit vector; a template guarantees two files differ only in the axes intentionally varied.

Axis	Name	Two levels
M	Metric	Vague (“judge appropriate”) vs. precise (CV accuracy + std)
B	Breadth	Stay in one model family vs. free to compare both
S	Stopping	Fixed budget of 3 vs. adaptive, up to 8
O	Output	Terse and numeric vs. fully reasoned prose
E	Emphasis	Explore broadly first vs. refine immediately

Step 3 — real experiments, not simulated ones

DESIGN AND EXECUTION

A resolution-III **Plackett–Burman design** selects 8 of the 32 possible configurations — enough to estimate every axis's individual effect without bias.

Every configuration runs against a real scikit-learn task: tune Random Forest / Gradient Boosting on the Breast Cancer Wisconsin dataset. Every number the agent sees is a genuine measurement, never invented.

8

configurations

18

total runs

46

real model fits

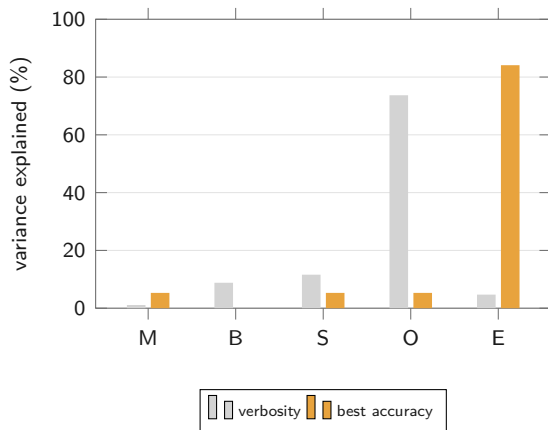
Step 4 — six numbers per run

EACH ANSWERS A SPECIFIC QUESTION

How much research happened	Did it pay off	Can the pipeline be trusted
■ # experiments run ■ # model families tried ■ search width (hyperparameter spread)	■ best cross-validated accuracy reached	■ verbosity (known ground-truth check) ■ distance to a reference research trace

Step 5 — which axis explains which behavior

SURROGATE MODEL RESULTS



Output format → verbosity

74% of variance, $p = 0.027$. Expected — confirms the pipeline works end to end.

Emphasis → accuracy

84% of variance. Refine-immediately beats explore-first by ~ 0.25 accuracy points — a non-obvious effect.

Step 6 — predict first, then run it for real

CAUSAL VALIDATION

One configuration was held out of the design; its prediction was written down before it was ever executed.

Number of experiments

PREDICTED
3.25

OBSERVED
3.00

Off by a quarter of an experiment — essentially exact.

Best accuracy reached

PREDICTED
0.9648

OBSERVED
0.9610

Off by ~ 0.004 — small, explainable miss.

A wrong-but-explainable prediction is more useful than a validation step that's skipped, or one that always happens to be right — it points straight at what the design couldn't see: an interaction between axes.

A white box — of behavior, not of mechanism

HOW MUCH OF THE BLACK BOX ACTUALLY OPENED

What became visible

- `program.md` became a labeled 5-axis vector, not an opaque file
- The axis $\rightarrow$ behavior link is a readable linear model, not another black box
- One prediction was checked against a real run before the outcome was known

What stayed opaque

- Nothing here explains the model's internal mechanism — weights, attention, computation
- Findings describe this study's own small system, not autoresearch loops in general
- Only the *method* is a candidate for generalizing — not yet the specific numbers

Limitations, stated plainly

READ THE RESULTS WITH THESE FIRMLY IN MIND

1. Agent = the author

Every run's decisions were made by someone who already knew this was an ablation study — a real confound.

2. Human trace is authored

Real experiment numbers, but an invented reasoning process — not an observed human.

3. Small sample

2 repeats/config; 8 configs → only 2 residual degrees of freedom in every model.

4. Narrow scope

One dataset, two model families, five axes — generalization not yet shown.

A method, demonstrated end to end

CONCLUSION & NEXT STEPS

Every step ran against real, executed machine-learning experiments — configuration space, ablation design, surrogate model, and a causal check that made a falsifiable prediction and tested it.

1. Replace the author-as-agent substitution with independent, fresh-context model calls
2. Record an actual human reference session to replace the authored trace
3. Expand the design (resolution IV+) to estimate the interaction Step 6 flagged
4. Repeat across more than one dataset to test generality

Full report: github.com/topoftheblock/autoresearch